SCIENCE OLYMPIAD at CORNELL

Water Quality

DIVISION C

INSTRUCTIONS

Please fill in your team number and your names before beginning this exam.
DO NOT open this exam until explicitly instructed to by the event supervisor.
You will have 50 minutes to complete this exam.
Communication with other teams is prohibited.

Team Number: _____

Names: _____

SCORE: _____

Tiebreaker (if needed) : WON | LOST

PLACE: _____

Multiple Choice

Select single best possible answer for the following questions, unless otherwise indicated (*1 point each).*

1. Which of these species are examples of marine phytoplankton?
    A. Dinoflagellates
    B. Diatoms
    C. Green algae
    D. Some cyanobacteria
    E. All of the above

2. Which *three* of these are methods of transportation of new nutrients into coastal ecosystems?
    A. Nitrogen fixation
    B. Regeneration
    C. Dry deposition
    D. Bioaccumulation
    E. Freshwater transport

3. The chief builders of coral reefs are which *two* of the following?
    A. Symbiotic fish
    B. Algae
    C. Nutrients
    D. Old Shells
    E. Stony Corals

4. Fish living in the epipelagic zone are likely to have which *three* of these characteristics?
    A. Streamlined or Fusiform Bodies
    B. Lunate Tail
    C. Narrow Caudal Peduncle
    D. Low muscle mass
    E. Silvery top, dark underside

5. If the phosphate level is too high, what might you expect to see?
    A. Biomagnification
    B. Eutrophication
    C. Stagnation
    D. Trophic Flagellation
    E. Acid Rain

6. Which of the following *two* tends to happen as a population approaches its carrying capacity?
    A. Increased competition for food
    B. Decreased competition for shelter
    C. Decreased competition for mating sites
    D. Increased competition for light
    E. All of the above

7. Coral bleaching is caused directly by:
    A. Global warming.
    B. The greenhouse effect.
    C. Coral losing their symbiotic algae.
    D. Coral living in older water.
    E. Tourists pouring bleach on the coral.

8. Which of the following *three* are adaptations seen in estuarine species?
    A. Increasing respiratory water flow
    B. Digging burrows
    C. Increased metabolic efficiency
    D. Shutting up shells
    E. None of the above

9. What are the three major ocean basins where most of the world's coral reefs are found?
    A. Atlantic, Southern, Pacific
    B. Indian, Pacific, Atlantic
    C. Southern, Atlantic, Indian
    D. Pacific, Arctic, Atlantic
    E. None of the above

10. To which of the following do corals belong to?
    A. Echinodermata
    B. Ctenophora
    C. Hemichordata
    D. Cnidaria
    E. All of the above

11. Ahermatypic corals typically:
    A. Have a symbiotic relationship with zooxanthellae
    B. Form small colonies
    C. Build reefs
    D. Reproduce sexually
    E. All of the above

12. Which of the following are correct?
    A. Water density is controlled more by temperature than salinity
    B. Water density is controlled more by salinity than temperature
    C. Water density is equally controlled by temperature and salinity
    D. Water density is not affected by temperature and salinity
    E. Water density is always constant

13. Temperature affects which *three* of the following?
    A. Phosphate levels
    B. Dissolved oxygen levels
    C. The rate of photosynthesis by aquatic plants
    D. The sensitivity of organisms to toxic wastes, parasites and diseases
    E. The salinity of an aquatic environment

14. Unpolluted waters usually have a nitrate level below what ppm?
    A. 4 ppm
    B. 3 ppm
    C. 2 ppm
    D. 1 ppm
    E. All of the above

15. Nitrogen occurs in water in which *three* of the following forms?
    A. Nitrate (NO_3)
    B. Nitrite (NO_2)
    C. Ammonia (NH_3)
    D. Ammonium (NH_4)
    E. Dinitrogen (N_2)

16. Turbid water may be the result of what factors?
    A. Soil erosion
    B. Urban run-off
    C. Algal blooms
    D. Bottom sediment disturbances which can be caused by boat traffic and abundant bottom feeders
    E. All of the above

True or False

Choose an answer of either True (A) or False (B) on your answer sheet for each question. Continue starting from question 17 on your answer sheet *(1 point each)*.

17. Lotic ecosystems are mainly flowing water.
18. Although untreated wastewater contains many billions of bacteria per gallon, most of these are not harmful to humans and some are even helpful and wastewater treatment.
19. Both nutrient cycles and energy flow are cyclic processes.
20. The total solids in wastewater are a combination of fixed solids and settleable solids.
21. Two types of autotrophs are photoautotrophs and bioautotrophs.
22. Excessive plant and algae growth create water quality problems.
23. Thermal pollution, the discharge of heated water from industrial operations, for example, can cause temperature changes that threaten the balance of aquatic systems.

Matching

Match each of the following terms to its description (*1 point each).*

Part I: Nutrients

24. Decomposition	a. The quantity of nutrients entering an ecosystem in a given period of time.
25. Nutrient Flux	b. The transfer of specific nutrient species from sediments to the overlying water (or from water to the sediments).
26. Diffusion	c. Releases carbon dioxide in the atmosphere from "once living" organisms.
27. Assimilation	d. Plants taking up nitrates in the soil and using it as a nutrient to build plant proteins.
28. Nutrient Loading	e. Process of carbon dioxide gas moving from air to water, or from water back to the air

Part II: Relationship Interactions

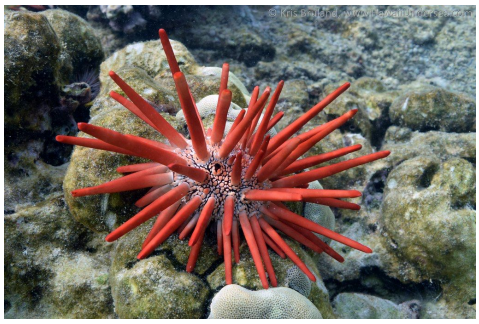
29. Parasitism	a. One species is killed, and the other benefits.
30. Commensalism	b. Interaction between two different organisms living in close physical association, typically to the advantage of both.
31. Predation	c. Both species benefit.
32. Mutualism	d. One species benefits and the other is neutral.
33. Symbiosis	e. One species is harmed but not killed, and the other benefits.

Part III: Ocean Zones

34. Abyssopelagic	A. The illuminated zone at the surface of the sea where there is enough light for photosynthesis. Nearly all primary production in the ocean occurs here.
35. Bathypelagic	B. Comprises of the water column that is near to (and is significantly affected by) the seabed and the benthos. Just above the benthic zone and forms a layer of the larger profundal zone.
36. Demersal	C. 1000-4000 meters down. Pitch black—no plant life. Most animals here survive by consuming the detritus falling from the zones above.
37. Epipelagic	D. 4000 meters to the ocean floor. Very few creatures live in the cold temperatures, high pressures and complete darkness of this depth.
38. Hadopelagic	E. Waters below 6000 meters, often include trenches and hydrothermal vents.

Image Identification:

A. 

B. 

C. 

D. 

E. 

F. 

G. 

H. 

39. Name species A: _____
    During what time of the day are they the most active predators? _____

40. Name species B: _____
    How do corals benefit from this species?

41. Name species C: _____
    Name the diet of this species:

42. Name species D: _____
    What happens if a male dies in a group?

43. Name species E: _____
    Describe how this organism contributes to bioerosion and the consequences.

44. Name species F: _____
    What adaptation do these fish have that allow them to consume hard shelled species?

45. Name species G: _____
    Name the effect of these species in coral reefs:

46. Name species H: _____
    Why is this organism a reef health indicator?

47. Which of the species above (A-H), if any, are indicators of overfishing?

48. Which of the species above (A-H), if any, are indicators of aquarium fish collection?

49. Which of the species above (A-H), if any, are indicators of nutrient pollution?

Fill in the Blank

carbonic acid	carbonate ions	hydrogen ions	bicarbonate
carbon dioxide	carboxylic acid	hydrogen carbonate	
calcium carbonate	calcium bicarbonate	hydroxide	

Ocean acidification occurs when CO2 dissolves in seawater, it forms **_50._**, which releases **_51._.** These ions then combine with **_52._** in the water to form **_53._**. Increasing the concentration of **_54._** in the atmosphere causes absorption in ocean water. Acidification decreases the availability of **_55._**, making it harder for corals and other organisms to form their shells.

Fill in the following table with the correct corresponding

Type of Water Pollution	Cause of Pollution	Effect of Pollution	Source of Pollution
Biodegradable waste	56.	60.	64.
Nutrients	57.	61.	65.
Heat	58.	62.	66.
Sedimentation	59.	63.	67.

Questions 56-59. Match the following **causes of pollution** with the type of water pollution in the table.

A. Suspended particles
B. Increased water temperature
C. Humans and animals
D. Nitrates and phosphate

Questions 60-63. Match the following **symptoms of pollution** with the type of water pollution in the table.

A. Decrease in oxygen, death of aquatic life
B. Increased temperature and turbidity
C. Eutrophication, algae blooms
D. Increased bacteria, decreased oxygen, death of aquatic life

Questions 64-67. Match the following **effects of pollution** with the type of water pollution in the table.

A. General runoff, effluent
B. Industrial runoff, wastewater treatment
C. Construction, farming, city runoff
D. Fertilizers

Short Answer

Please write the answers on the answer sheet provided. Work written on these pages will be ignored.

68. The accumulation and deposition of what two things creates coral reefs?

69. The first stage of the coral's life cycle is called the _____, which is free swimming.

70. The second stage of the life cycle of coral is the _____, when the coral is sedentary.
    a. In this stage, the coral is able to reproduce (circle one): T / F

71. Provide the correct order of the following events in sexual reproduction of corals: larvae settlement, spawning, fertilizing, cloning. (*1 point)*

72. Name five major requirements for reef formation *(1 point per each correct example listed. If there are more than 5 answers, only the first 5 will be considered):*

73. Name five major coral reef threats *(1 point per each correct example listed. If there are more than 5 answers, only the first 5 will be considered):*

74. List three examples of biogeochemical cycles *(1 point per each correct example listed. If there are more than 3 answers, only the first 3 will be considered).*

75. List three watershed resource management issues *(1 point per each correct example listed. If there are more than 3 answers, only the first 3 will be considered).*

76. List three effects of thermal pollution *(1 point per each correct example listed. If there are more than 3 answers, only the first 3 will be considered).*